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Lab Report: RF Amplifier Characterization

Experiments performed (Part 2): Gain vs Frequency (D1), Harmonics and 1-dB Compression Point (D2), Third-order Intercept (TOI) (D3).

1. Objective

The objective of this experiment is to characterize the RF amplifier used in the receiver chain. Specifically, this report covers:

- To measure the amplifier gain as a function of frequency.
- To measure the amplifier gain compression with increasing input power and determine the 1-dB compression point.
- To measure the amplifier's third-order intermodulation products using a two-tone test and determine the third-order intercept (TOI) point.

2. Theory

2.1 RF Amplifier Gain and Harmonics

An RF amplifier increases the power of an input signal. Its small-signal gain is:

$$\text{Gain} = P_{out} - P_{in}$$

A nonlinear amplifier can be represented as a power-series expansion of its input:

$$y(t) = a_1x(t) + a_2x^2(t) + a_3x^3(t) + ...$$

For a single-tone input $x(t) = A\cos(\omega t)$, the output contains a DC term, the fundamental at ω, and harmonics at 2ω, 3ω, and higher frequencies. At sufficiently high input power, gain compression occurs and the amplifier gain decreases. The 1-dB compression point is defined as the value of P_{in} at which the gain has fallen by 1 dB from the small-signal gain; the corresponding output power is denoted $P_{out}(1dB)$.

2.2 Third-order Intercept (TOI)

When two closely spaced input tones at f_1 and f_2 are applied to a nonlinear amplifier, the output contains the fundamentals f_1 and f_2, second-order products, harmonics, and third-order intermodulation (IM3) products. The two IM3 products closest to the fundamentals are:

$$2f_1 - f_2 \text{ and } 2f_2 - f_1$$

The fundamental output power increases with input power at a slope of approximately 1:1 (on a dB–dB scale), while the IM3 output increases at a slope of approximately 3:1. The extrapolated intersection of these two trends is called the third-order intercept point (TOI). In practice the amplifier saturates before this point is physically reached, so it is obtained only by extrapolation. Using the ideal 1:1 and 3:1 slopes, the input- and output-referred intercept points can be computed directly from a single measurement at power level Pin as:

$$IIP3 = Pin + (P_{out,fund} - P_{out,3rd}) / 2,\quad OIP3 = P_{out,fund} + (P_{out,fund} - P_{out,3rd}) / 2$$

Where “fund” indicates “fundamental”.

3. Components Required

- RF signal generator(s)
- Spectrum analyser
- RF amplifier and the required DC power supply
- RF power combiner (for the two-tone TOI test)
- 50 Ω terminations
- Suitable RF cables, adapters, and attenuators

4. Experimental Setup

4.1 D1 - Gain vs Frequency

The amplifier was powered using a DC supply of +18 V. Setup A (signal generator → amplifier → spectrum analyser) was used. The generator power was held fixed at Pin = −10 dBm while its frequency was swept from 50 MHz to 2000 MHz, and the output power was recorded on the spectrum analyser at each frequency.

4.2 D2 - Harmonics and 1-dB Compression Point

The generator frequency was fixed at 50 MHz. Starting from a low input power, P_{in} was increased in 1 dB steps from −20 dBm to −3 dBm, and the corresponding output power P_{out} was recorded at each step using Setup A.

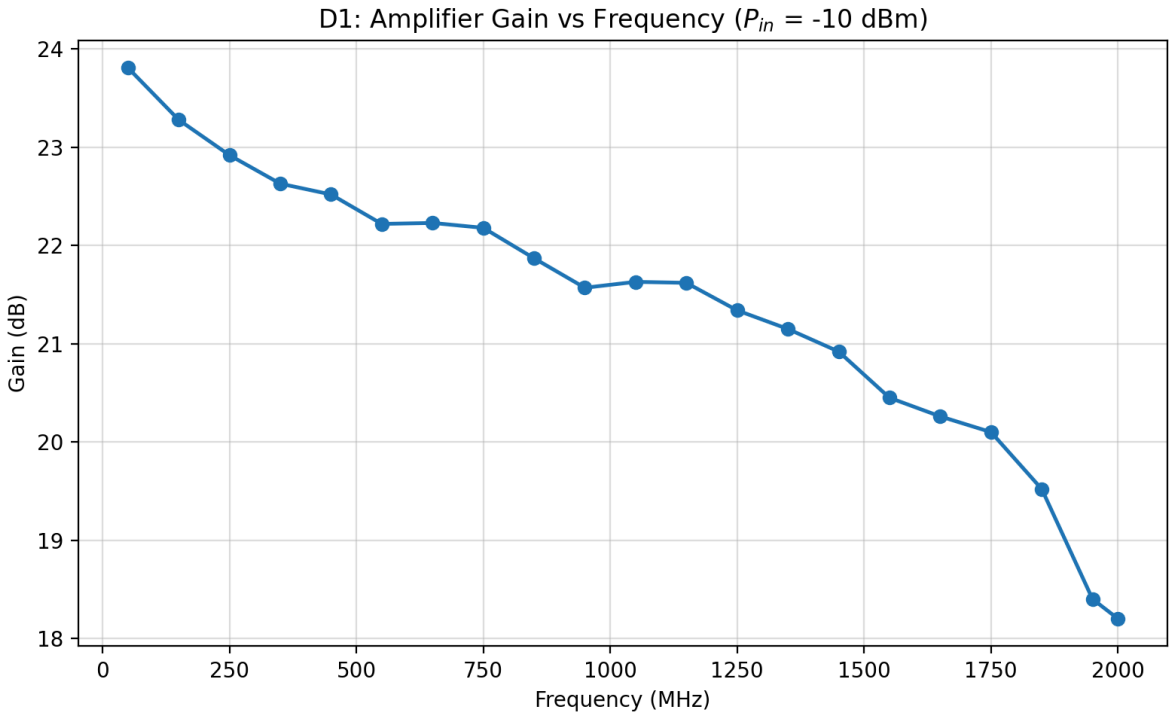
4.3 D3 - Third-order Intercept

Two signal generators were set to f_1 = 13 MHz and f_2 = 10 MHz, giving third-order intermodulation products at $2f_2 - f_1$ = 7 MHz and $2f_1 - f_2$ = 16 MHz. The two generator outputs were combined using the RF power combiner and applied to the amplifier input, with the amplifier output connected to the spectrum analyser. The power of f_2 was held fixed at −10 dBm, while the power of f_1 (P_{in} per tone) was varied from −10 dBm to −5 dBm. At each step, the fundamental output power and the output power of the $2f_1 - f_2$ (16 MHz) third-order product were recorded.

5. Observations and Graphs

5.1 D1 - Gain vs Frequency (Pin = −10 dBm)

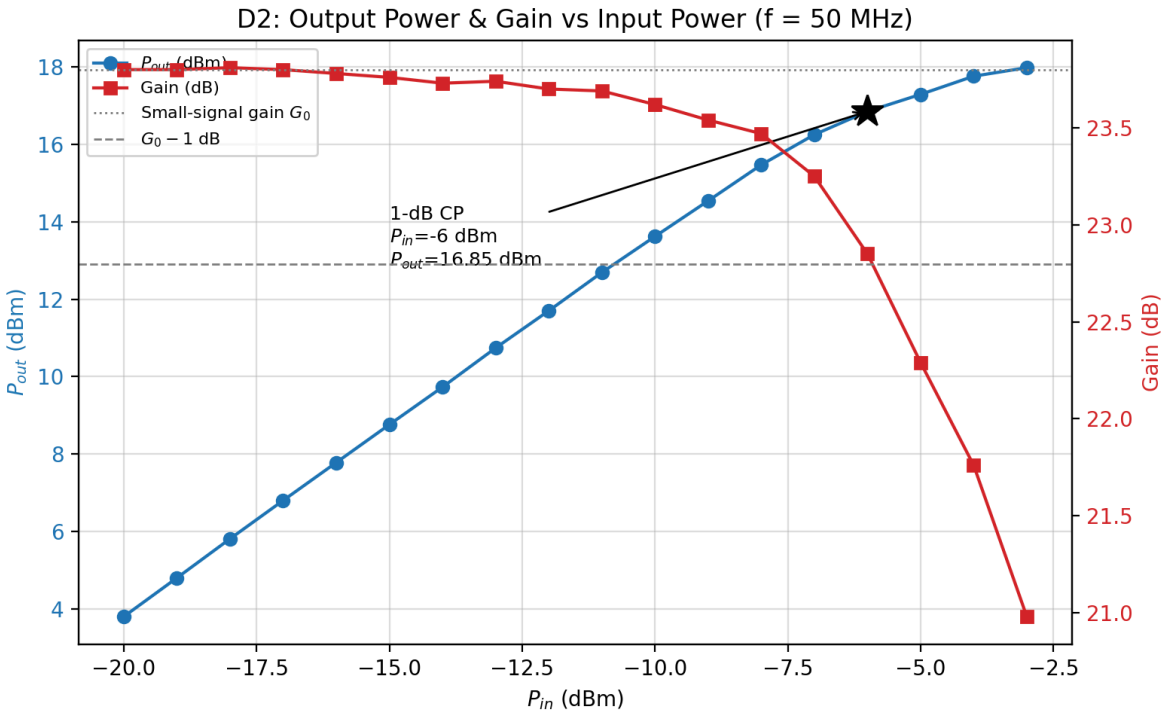
Frequency (MHz)	Pin (dBm)	Pout (dBm)	Gain (dB)
50	-10.00	13.81	23.81
150	-10.00	13.28	23.28
250	-10.00	12.92	22.92
350	-10.00	12.63	22.63
450	-10.00	12.52	22.52
550	-10.00	12.22	22.22
650	-10.00	12.23	22.23
750	-10.00	12.18	22.18
850	-10.00	11.87	21.87
950	-10.00	11.57	21.57
1050	-10.00	11.63	21.63
1150	-10.00	11.62	21.62
1250	-10.00	11.34	21.34
1350	-10.00	11.15	21.15
1450	-10.00	10.92	20.92
1550	-10.00	10.45	20.45
1650	-10.00	10.26	20.26
1750	-10.00	10.10	20.10
1850	-10.00	9.52	19.52
1950	-10.00	8.40	18.40
2000	-10.00	8.20	18.20



5.2 D2 - Harmonics and 1-dB Compression (f = 50 MHz)

Pin (dBm)	Pout (dBm)	Gain (dB)
-20.00	3.80	23.80
-19.00	4.80	23.80
-18.00	5.81	23.81
-17.00	6.80	23.80
-16.00	7.78	23.78
-15.00	8.76	23.76
-14.00	9.73	23.73
-13.00	10.74	23.74
-12.00	11.70	23.70
-11.00	12.69	23.69
-10.00	13.62	23.62
-9.00	14.54	23.54
-8.00	15.47	23.47
-7.00	16.25	23.25
-6.00	16.85	22.85 (the 1 dB compression point)

-5.00	17.29	22.29
-4.00	17.76	21.76
-3.00	17.98	20.98

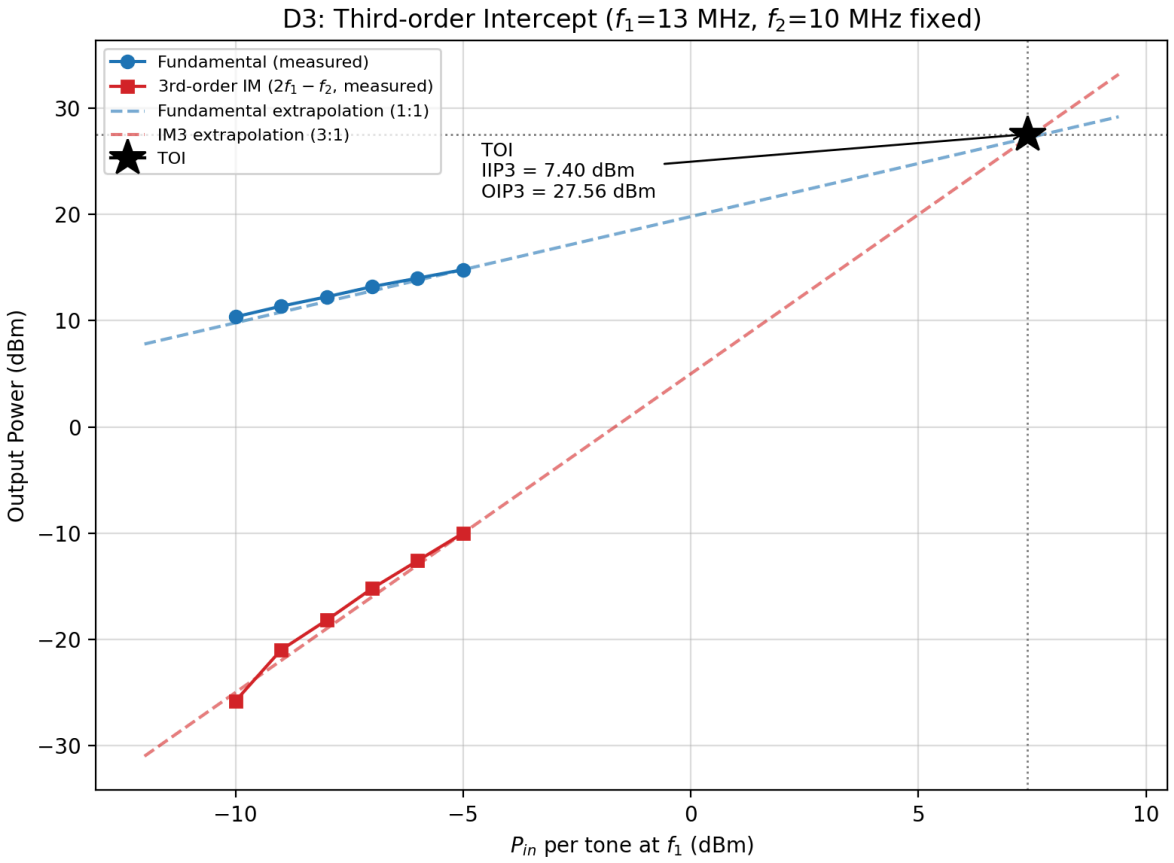


The blue line depicts P_{out} (dBm), and the red line shows Gain (dB).

For example, at P_{in} = -20 dBm, the P_{out} is 3.80 dBm (the blue dot), and Gain is 23.8 dB, the red square.

5.3 D3 - Third-order Intercept (f_1 = 13 MHz, f_2 = 10 MHz, f_2 is fixed, only P_{in} of f1 is varied)

P_{in} per tone at f_1 (dBm)	P_{out} fundamental (dBm)	P($2f_1-f_2$) 3rd-order (dBm)
-10.00	10.36	-25.80
-9.00	11.36	-21.00
-8.00	12.24	-18.20
-7.00	13.20	-15.21
-6.00	14.00	-12.60
-5.00	14.80	-10.00



6. Calculations and Results

6.1 D1 - Gain vs Frequency

Gain was calculated at each frequency as Gain = $P_{out} - P_{in}$, with P_{in} = −10 dBm fixed. The gain is highest at low frequency (23.81 dB at 50 MHz) and falls off steadily with increasing frequency, dropping to 18.20 dB by 2 GHz. This roll-off is the expected behaviour of a practical RF amplifier, whose gain is not flat indefinitely but decreases as the operating frequency approaches and exceeds the upper edge of its designed operating band.

6.2 D2 - 1 dB Compression Point

The small-signal gain, averaged over the least-compressed low-power points (P_{in} = −20 to −16 dBm), is:

$$G_0 = 23.8 \text{ dB}$$

The gain at each P_{in} was compared against the target $G_0 - 1 = 22.8$ dB. The measured gain crosses this threshold at P_{in} = −6 dBm, where Gain = 22.85 dB (only 0.05 dB above the 1-dB-down target, the closest point in the swept data). The 1-dB compression point is therefore taken as:

$$P_{in}(1\text{dB}) \approx -6 \text{ dBm}, \; P_{out}(1\text{dB}) \approx 16.85 \text{ dBm}$$

Beyond this point the gain continues to fall rapidly (down to 20.98 dB by Pin = −3 dBm), confirming the amplifier is well into compression at higher drive levels.

6.3 D3 - Third-order Intercept (TOI)

For each measured point, IIP3 and OIP3 were calculated using the ideal-slope formula:

IIP3 = Input-referred Third-order Intercept Point

OIP3 = Output-referred Third-order Intercept Point

$$IIP3 = Pin + (Pout,fund - Pout,3rd)/2,\ OIP3 = Pout,fund + (Pout,fund - Pout,3rd)/2$$

Where “fund” indicates “fundamental”.

The calculated values at each input power were:

P_{in} (dBm)	IIP3 (dBm)	OIP3 (dBm)
-10	8.08	28.44
-9	7.18	27.54
-8	7.22	27.46
-7	7.21	27.41
-6	7.30	27.30
-5	7.40	27.20

Averaging over all six points gives the reported third-order intercept:

$$IIP3 \approx 7.40 \text{ dBm},\ OIP3 \approx 27.56 \text{ dBm}$$

As a cross-check, a linear regression fit to the measured data gives a fundamental slope of ≈0.89:1 (close to the ideal 1:1) and an IM3 slope of ≈3.06:1 (close to the ideal 3:1), confirming genuine third-order intermodulation behaviour. The fundamental slope being slightly below 1 reflects that mild gain compression has already begun over this power range, consistent with the compression already observed in D2 at similar power levels.

7. Conclusion

Summary of key results:

- Amplifier small-signal gain ≈ 23.8 dB at 50 MHz, falling to ~18.2 dB by 2 GHz.
- 1-dB compression point: $P_{in} \approx -6$ dBm, $P_{out} \approx 16.85$ dBm.
- Third-order intercept: IIP3 ≈ 7.4 dBm, OIP3 ≈ 27.6 dBm, with measured fundamental/IM3 slopes of ≈0.89:1 and ≈3.06:1, consistent with the ideal 1:1 and 3:1 behaviour.

Learnings

This set of experiments extended what I had learnt earlier about individual RF components into a fully working system and gave me a much better feel for how gain, compression, and distortion are actually characterized in practice. In D1 I saw that amplifier gain is not flat with frequency; it is highest near the low end of the band and rolls off as frequency increases. D2 taught me what gain compression really means: as long as the input power is low, the gain stays essentially constant, but past a certain drive level the amplifier can no longer scale its output linearly, and the 1-dB compression point is just the input power where that departure first becomes clearly visible. D3 built on this using a two-tone test and showed me why third-order products matter: because they grow three times as fast (in dB) as the fundamental, they eventually catch up to it, and the extrapolated crossing point (the TOI) gives a single number that summarizes how linear an amplifier really is, even though the amplifier itself saturates long before that point is physically reached.

Beyond the formal D1–D3 measurements, we also built the full chain: signal generator, power splitter, power combiner, amplifier, and spectrum analyser as a system check. Splitting a 0 dBm input should ideally give −3 dBm at each output, but I measured a slightly larger loss, and recombining the two halves should ideally restore 0 dBm but instead gave about −2 dBm, showing the splitter and combiner each add their own real loss on top of the ideal 3 dB. Feeding this into the amplifier and reading the output on the spectrum analyser let me calculate a system gain of about 20 dB, which matched closely with the ≈20.98 dB gain read directly from the D2 data at a similar input power, and the same check at a much lower drive level (−20 dBm into the system) again matched the D2 gain at −20 dBm. Getting the same gain number from two completely different measurement paths was a good practical lesson in how to verify that an entire instrument chain is behaving correctly, and it also reinforced why keeping the two combiner input cables equal in length matters: unequal lengths would introduce a phase difference between the two paths and corrupt the combined signal before it even reached the amplifier.

Derivation of IIP3 and OIP3

At a two-tone measurement with input power P_{in}, the measured outputs are:
- Fundamental output: Pout,fund
- Third-order IM product output: Pout,3rd

On a dB–dB plot:
- The fundamental line has slope 1.
- The IM3 line has a slope of 3.

Step 1: Write each line's equation.

Let x be input power, and let d = x − Pin.

Fundamental: y_fund(x) = Pout,fund + d
IM3: y_3rd(x) = Pout,3rd + 3d

Step 2: Set them equal (this defines the TOI).

$P_{out,fund} + d = P_{out,3rd} + 3d$

Step 3: Solve for d.

$P_{out,fund} - P_{out,3rd} = 2d$

$d = (P_{out,fund} - P_{out,3rd}) / 2$

Step 4: IIP3 (x-coordinate of intersection)

Since $x = P_{in} + d$:

$IIP3 = P_{in} + (P_{out,fund} - P_{out,3rd}) / 2$

Step 5: OIP3 (y-coordinate of intersection)

$OIP3 = P_{out,fund} + d$

$OIP3 = P_{out,fund} + (P_{out,fund} - P_{out,3rd}) / 2$

Result

$IIP3 = P_{in} + (P_{out,fund} - P_{out,3rd}) / 2$
$OIP3 = P_{out,fund} + (P_{out,fund} - P_{out,3rd}) / 2$